Statistics Notes

A Deep Dive into Statistics and Statistical Inference and Learning

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1 Fundamental Reminders

Covariance & Variance

Covariance:

$$\mathrm{Cov}(X,Y) = E((X-\mu_X)(Y-\mu_Y))$$
$$\Leftrightarrow_{\text{Theorem}} \mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

Variance Reminder:

$$E\left[(X-\mu)^2\right] = E(X^2) - E(X)^2$$

Variance-Covariance Matrix

$$V(X) = \begin{pmatrix} C(X_1,X_1) & C(X_1,X_2) & \cdots & C(X_1,X_k) \\ C(X_2,X_1) & C(X_2,X_2) & \cdots & C(X_2,X_k) \\ \vdots & \vdots & \ddots & \vdots \\ C(X_k,X_1) & C(X_k,X_2) & \cdots & C(X_k,X_k) \end{pmatrix}$$

where $X = \begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix}$ and $C(X_i,X_i) = V(X_i)$.

Conditional Expectations

$$E(X \mid Y=y) = \begin{cases} \sum x f_{X|Y}(x \mid y)\,dx & \text{(Discrete)} \\ \int x f_{X|Y}(x \mid y)\,dx & \text{(Continuous)} \end{cases}$$

where:

- $f_{X|Y}(x \mid y) \overset{\text{(Discrete)}}{=} P(X=x \mid Y=y) = \frac{P(X=x,Y=y)}{P(Y=y)} = \frac{f_{X,Y}(x,y)}{f_Y(y)}$
- $f_{X|Y}(x \mid y) \overset{\text{(Continuous)}}{=} \frac{f_{X,Y}(x,y)}{f_Y(y)}, \quad f_Y(y) > 0$

Properties of $f_{X,Y}$: $f_{X,Y}$ is $P(X=x,Y=y)$ if X,Y are discrete. Else, it is such that:

(i) $f_{X,Y}(x,y) \geq 0 \quad \forall (x,y)$

(ii) $\int_{\mathbb{R}} \int_{\mathbb{R}} f_{X,Y}(x,y)\,dx\,dy = 1$

(iii) $\forall A \subset \mathbb{R}^2 \left(P((X,Y) \in A) = \iint_A f_{X,Y}\right)$

One also benefits from reminding the CLT:

Theorem: Central Limit Theorem

Let $\{X_i\}_{i=1}^{\infty}$ be iid with mean μ and variance σ^2. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then,

$$Z_n := \frac{\overline{X}_n - \mu}{\sqrt{V(\overline{X}_n)}} = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \rightsquigarrow Z \sim N(0,1)$$

One can ask: *"How accurate are the approximations given by the CLT?"*

Theorem: Berry-Esseen Inequality

Suppose $E|X_1^3| < \infty$. Then

$$\sup_z |P(Z_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{E|X_1 - \mu|^3}{\sqrt{n}\cdot\sigma^3}$$

If Y_n has a limiting Normal distribution, the following method allows us to find the limiting distribution of $g(Y_n)$ where g is any smooth function.

Theorem: Delta Method

Suppose that $\frac{\sqrt{n}(Y_n-\mu)}{\sigma} \rightsquigarrow N(0,1)$ and that g is a differentiable function such that $g'(\mu) \neq 0$. Then $\frac{\sqrt{n}(g(Y_n)-g(\mu))}{|g'(\mu)|\sigma} \rightsquigarrow N(0,1)$. In other words,

$$Y_n \approx N(\mu, \sigma^2/n) \implies g(Y_n) \approx N\left(g(\mu), g'(\mu)^2\frac{\sigma^2}{n}\right)$$

2 Models, Statistical Inference and Learning

Statistical inference, or learning, as it is called in computer science, is the process of using data to infer the distribution that generated the data.

The typical question in this field of study would be: Given $X_1, \ldots, X_n \sim F$, a sample, how do we infer F. In some cases we might just infer some feature of F such as its mean.

2.1 Parametric and Nonparametric Models

Statistical Model

A **statistical Model** $\mathcal{F}$ is a set of distributions (or densities, or regression functions).

Parametric Model

A **parametric model** is a set $\mathcal{F}$ that can be parametrized by a finite number of parameters.

Example: Normal Distribution Model

Assuming the data follows a Normal distribution, one model is:

$$\mathcal{F} = \left\{ f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2 \right), \mu \in \mathbb{R}, \sigma > 0 \right\}$$

Such model is a two-parameter model. In general, a parametric model has the form

$$\mathcal{F} = \{ f(x; \theta) \mid \theta \in \Theta \}$$

where θ is an unknown parameter (or vector of such object) that take values in Θ. If $\theta = (\theta_1, \ldots, \theta_n)$ and we are only interested in one component, say θ_i, then the remaining parameters are called **nuisance parameters**.

Nonparametric Model

A **nonparametric model** is a set $\mathcal{F}$ that cannot be parametrized by a finite number of parameters. For example, $\mathcal{F}_{ALL} = \{\text{all CDF's}\}$ is nonparametric.

Example: Statistical Functional

Let $X_1, \ldots, X_n \sim F$. Suppose that we mean to estimate $\mu = E(X_1) = \int x\, dF(x)$ assuming only that μ exists. The mean μ may be thought of as a function of F: $\mu = T(F) = \int x\, dF(x)$. In general, any function of F is called a **statistical functional**.

Other examples would be the variance, $T(F) = \int x^2\, dF(x) - \left(\int x\, dF(x) \right)^2$, and the median, $T(F) = F^{-1}(1/2)$.

Example: Regression Models

Suppose we observe pairs of data $(X_i, Y_i)_{i=1}^n$. Perhaps X_i is the blood pressure and Y_i is how long the subject lived. X is called a **predictor** (or regressor). Y is called the **outcome** (or the response variable).
We call $r(x) = E(Y \mid X = x)$ the **regression function**.

- If we assume that $r \in \mathcal{F}$ where $\mathcal{F}$ is finite-dimensional, then we have a **parametric regression model**.
- If we assume the opposite, we have a **nonparametric regression model**.

Regression models are often written as:

$$Y = r(X) + \epsilon \quad \text{where } E(\epsilon) = 0$$

Actually, one can always rewrite a regression model this way:

$$[\epsilon = Y - r(X) \implies Y = Y - r(X) + r(X) = r(X) + \epsilon]$$

$$\text{And } E(\epsilon) = E(E(\epsilon \mid X)) = E(E(Y - r(X) \mid X)) = E(r(X) - r(X)) = 0.$$

Notation

If $\mathcal{F} = \{f(x;\theta) \mid \theta \in \Theta\}$ is a parametric model, we write $P_\theta(X \in A) = \int_A f(x;\theta)\,dx$ and $E_\theta(r(X)) = \int r(x) f(x;\theta)\,dx$. Similarly, $V_\theta = E_\theta(r(X)^2) - E_\theta(r(X))^2$.

2.2 Fundamental Concepts in Inference

Many (but not all) inferential problems can be identified as one of three types:

1. Estimation
2. Confidence Sets
3. Hypothesis Testing

2.3 Point Estimation

Point estimation refers to providing a single "best guess" of some quantity of interest. The quantity of interest could be a parameter in a parametric model, a CDF, a PDF, a regression function, or a prediction for a future value Y.

By convention, we denote a point estimate of θ by $\hat{\theta}$ or $\hat{\theta}_n$. Remember that θ is a fixed, unknown quantity. The estimate $\hat{\theta}$ depends on the data, so $\hat{\theta}$ is a **random variable**.

Estimator & Bias

Let $X_1, \ldots, X_n$ be n IID data points from some distribution F. A point estimator $\hat{\theta}_n$ of θ is some function of $X_1, \ldots, X_n$: $\hat{\theta}_n = g(X_1, \ldots, X_n)$.

The **bias** of an estimator is defined by:

$$\text{bias}(\hat{\theta}_n) = E_\theta(\hat{\theta}_n) - \theta$$

We say that $\hat{\theta}_n$ is **unbiased** if $\text{bias}(\hat{\theta}_n) = 0$.

Consistency

A point estimator $\hat{\theta}_n$ of a parameter θ is **consistent** if $\hat{\theta}_n \xrightarrow{P} \theta$.

The distribution of $\hat{\theta}_n$ is called the **sampling distribution**. The standard deviation of $\hat{\theta}_n$ is called the **standard error** ($\text{se} \equiv \text{se}(\hat{\theta}_n) := \sqrt{V(\hat{\theta}_n)}$). Often, the se depends on the unknown F. The estimated se is denoted $\hat{\text{se}}$.

Example: Bernoulli Estimator

Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and let $\hat{p}_n = n^{-1}\sum_{i=1}^n X_i$. Then $E(\hat{p}_n) = n^{-1}\sum_{i=1}^n E(X_i) = p$, so $\hat{p}_n$ is unbiased.
The standard error is $\text{se} = \sqrt{V(\hat{p}_n)} = \sqrt{p(1-p)/n}$. Hence, $\hat{\text{se}} = \sqrt{\hat{p}_n(1-\hat{p}_n)/n}$.

The quality of a point estimate is sometimes assessed by the **mean squared error** (MSE):

$$\text{MSE} = E_\theta((\hat{\theta}_n - \theta)^2)$$

Theorem

$\text{MSE} = \text{bias}^2(\hat{\theta}_n) + V_\theta(\hat{\theta}_n)$.

Proof. Let $\overline{\theta}_n = E_\theta(\hat{\theta}_n)$. Then

$$\begin{aligned}
\text{MSE} &= E_\theta((\hat{\theta}_n - \theta)^2) \\
&= E_\theta((\hat{\theta}_n - \overline{\theta}_n + \overline{\theta}_n - \theta)^2) \\
&= E_\theta((\hat{\theta}_n - \overline{\theta}_n)^2) + (\overline{\theta}_n - \theta)^2 \\
&= V_\theta(\hat{\theta}_n) + \text{bias}^2(\hat{\theta}_n)
\end{aligned}$$

□

Theorem

If $\text{bias} \to 0$ and $\text{se} \to 0$ as $n \to \infty$ then $\hat{\theta}_n$ is consistent, that is, $\hat{\theta}_n \xrightarrow{P} \theta$.

Proof. $(\text{bias} \to 0, \text{se} \to 0) \implies \text{MSE} \to 0 \implies \hat{\theta}_n \xrightarrow{mq} \theta \implies \hat{\theta}_n \xrightarrow{P} \theta$. □

Asymptotically Normal

An estimator is **asymptotically Normal** if $\frac{\hat{\theta}_n - \theta}{\text{se}} \rightsquigarrow N(0,1)$.

2.4 Confidence Sets

Confidence Interval

A $1-\alpha$ **confidence interval** for a parameter θ is an interval $C_n = (a,b)$, where $a = a(X_1,\ldots,X_n)$ and $b = b(X_1,\ldots,X_n)$ are functions such that

$$P_\theta(\theta \in C_n) \geq 1-\alpha, \quad \forall \theta \in \Theta$$

We call $1-\alpha$ the **coverage** of this interval.

Example: Bernoulli Confidence Interval

Again, one falls into the Bernoulli setting. Let $C_n = (\hat{p}_n - \varepsilon_n, \hat{p}_n + \varepsilon_n)$, where $\varepsilon_n^2 = \log(2/\alpha)/2n$. By applying Hoeffding's inequality,

$$P(p \in C_n) \geq 1-\alpha, \quad \forall p \in [0,1].$$

Hence, C_n is a $1-\alpha$ confidence interval.

Theorem

Suppose that $\hat{\theta}_n \approx N(\theta, \hat{\text{se}}^2)$. Let Φ be the CDF of a standard Normal and let $z_{\alpha/2} = \Phi^{-1}(1-(\alpha/2))$. Let $C_n = (\hat{\theta}_n - z_{\alpha/2}\hat{\text{se}}, \hat{\theta}_n + z_{\alpha/2}\hat{\text{se}})$. Then

$$P_\theta(\theta \in C_n) \to 1-\alpha \quad (\text{as } n \to \infty).$$

Proof. Let $Z_n := (\hat{\theta}_n - \theta)/\hat{\text{se}}$. By assumption, $Z_n \rightsquigarrow Z \sim N(0,1)$. Hence,

$$\begin{aligned} P_\theta(\theta \in C_n) &= P_\theta\left(-z_{\alpha/2} < \frac{\hat{\theta}_n - \theta}{\hat{\text{se}}} < z_{\alpha/2}\right) \\ &\to P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1-\alpha \end{aligned}$$

□

2.5 Hypothesis Testing

In hypothesis testing, we start with some default theory, the **null hypothesis**, which is denoted H_0, and we ask:

Q.: *With this data, do we have enough evidence to reject my theory?*

If not, we retain/accept the null hypothesis.

We will follow as follows. **Chapter Sequence Diagram:**

